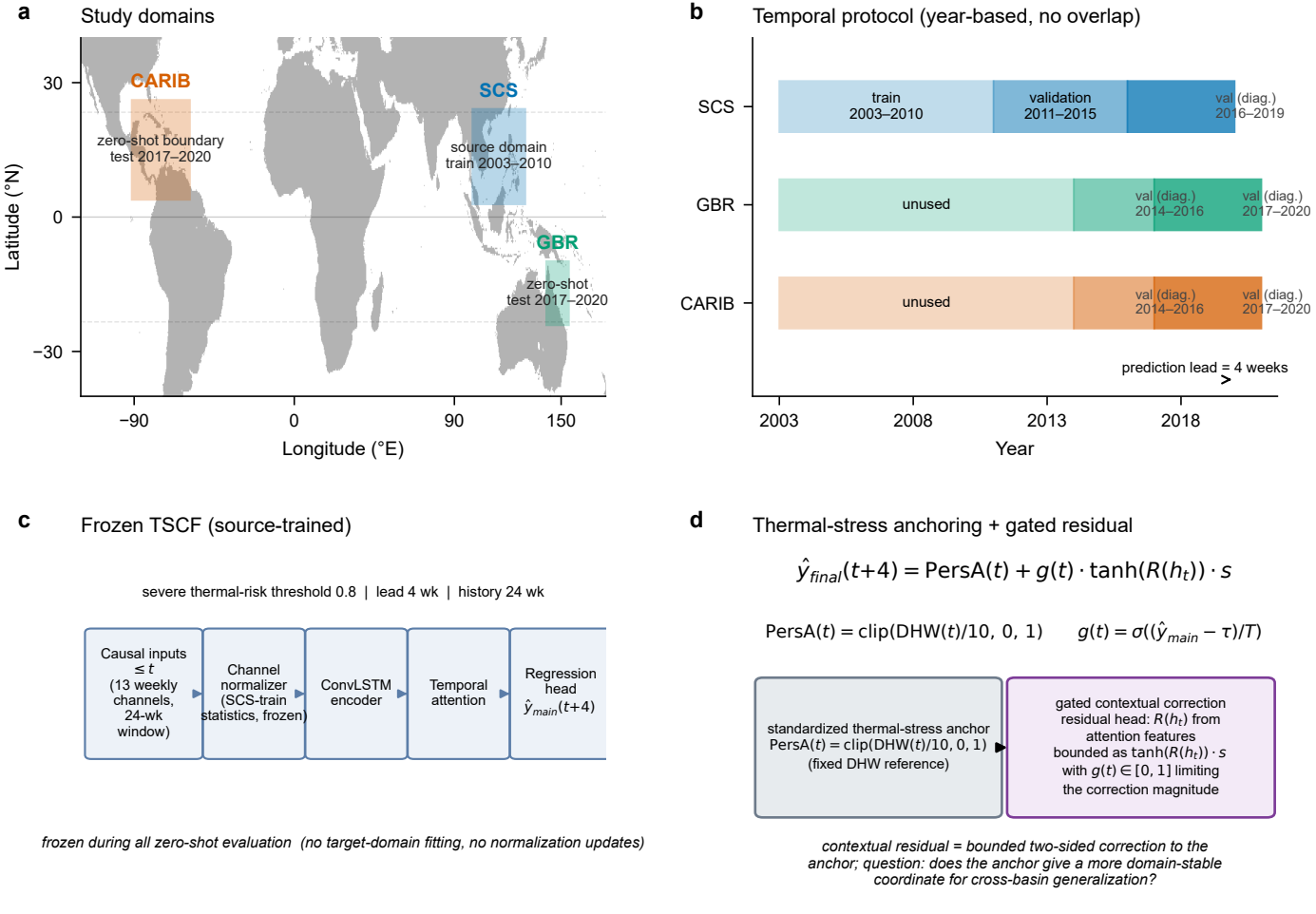

a
Study domains
CARIB
zero-shot boundary
test 2017–2020
SCS
source domain
train 2003–2010
GBR
zero-shot
test 2017–2020
30
0
−30
Latitude (°N)
−90
0
90
150
Longitude (°E)
b
Temporal protocol (year-based, no overlap)
SCS
train
2003–2010
validation
2011–2015
val (diag.)
2016–2019
GBR
unused
val (diag.)
2014–2016
val (diag.)
2017–2020
CARIB
unused
val (diag.)
2014–2016
val (diag.)
2017–2020
prediction lead = 4 weeks
2003
2008
2013
2018
Year
c
Frozen TSCF (source-trained)
severe thermal-risk threshold 0.8 | lead 4 wk | history 24 wk
Causal inputs
≤ t
(13 weekly
channels,
24-wk
window)
Channel
normalizer
(SCS-train
statistics, frozen)
ConvLSTM
encoder
Temporal
attention
Regression
head
ŷ_main(t+4)
frozen during all zero-shot evaluation (no target-domain fitting, no normalization updates)
d
Thermal-stress anchoring + gated residual
ŷ_final(t+4) = PersA(t) + g(t) · tanh(R(h_t)) · s
PersA(t) = clip(DHW(t)/10, 0, 1)
g(t) = σ((ŷ_main − τ)/T)
standardized thermal-stress anchor
PersA(t) = clip(DHW(t)/10, 0, 1)
(fixed DHW reference)
gated contextual correction
residual head: R(h_t) from
attention features
bounded as tanh(R(h_t)) · s
with g(t) ∈ [0, 1] limiting
the correction magnitude
contextual residual = bounded two-sided correction to the
anchor; question: does the anchor give a more domain-stable
coordinate for cross-basin generalization?